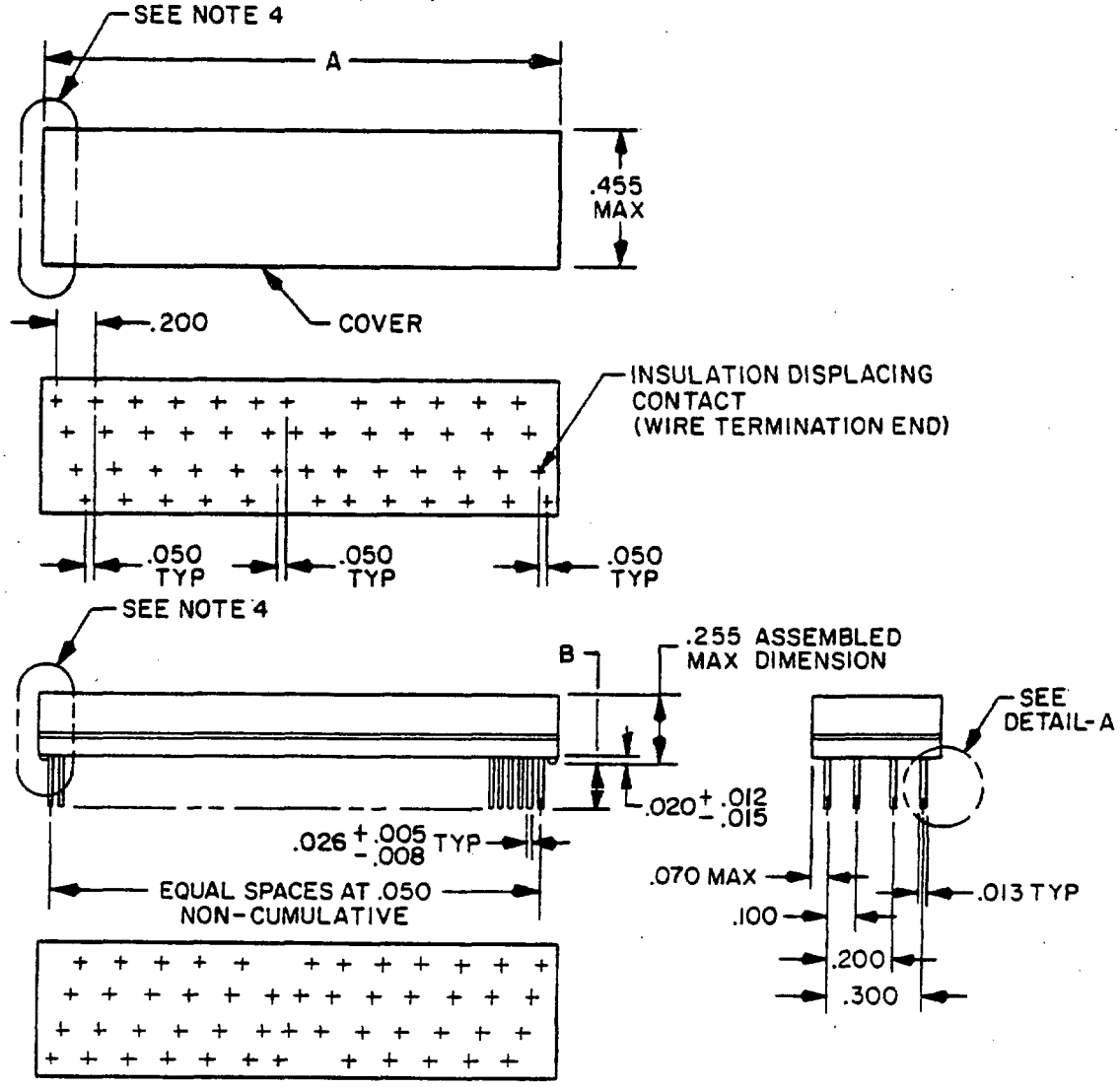
MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, FLAT CABLE, NONENVIRONMENTAL, PRINTED WIRING BOARD SOLDER TRANSITION FOR TERMINATING ROUND CONDUCTORS ON .050 INCH CENTERS

This specification sheet is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the PWB connectors described herein shall consist of this specification sheet and the latest issue of MIL-C-83503(USAF).

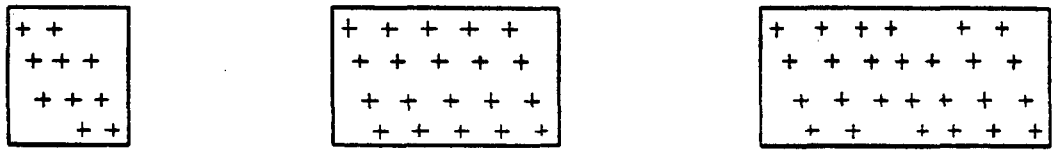


-11 & -12
(SEE PAGE 2 FOR -01 THRU -10)

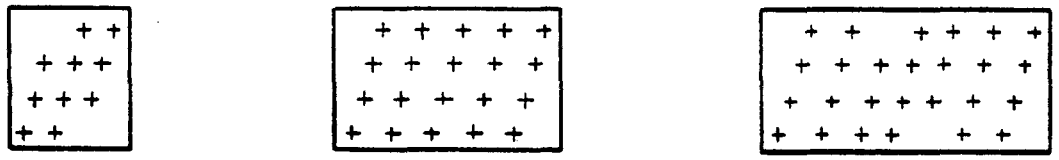
FIGURE 1. Dimensions and configuration.

FSC 5935

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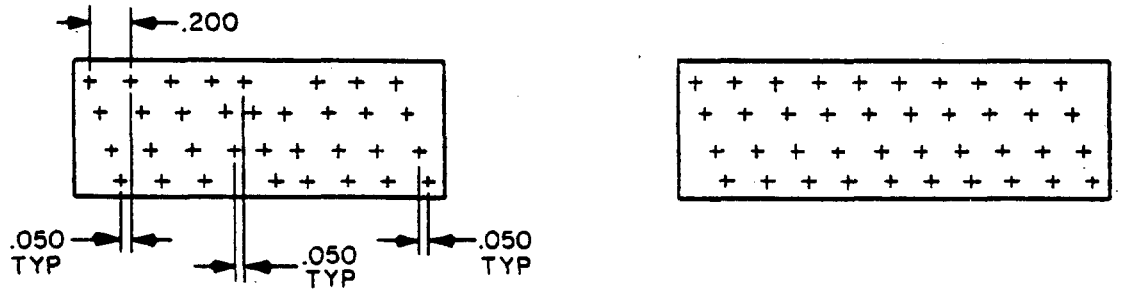


TOP VIEWS

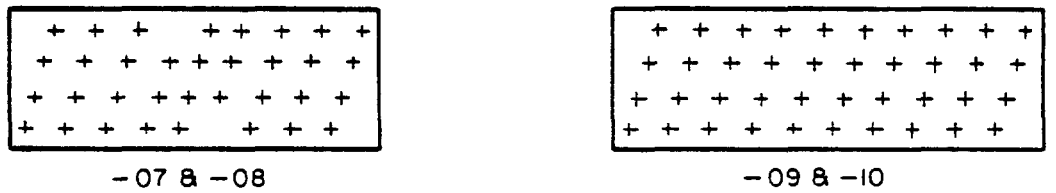


-01 & -02 -03 & -04 -05 & -06

BOTTOM VIEWS



TOP VIEWS

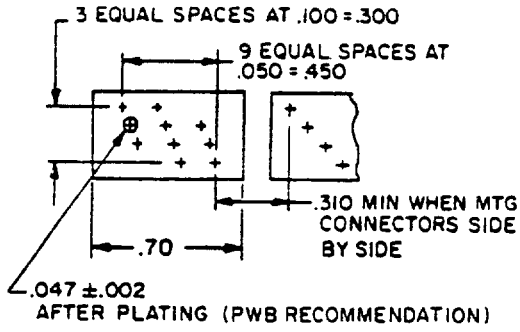


-07 & -08 -09 & -10

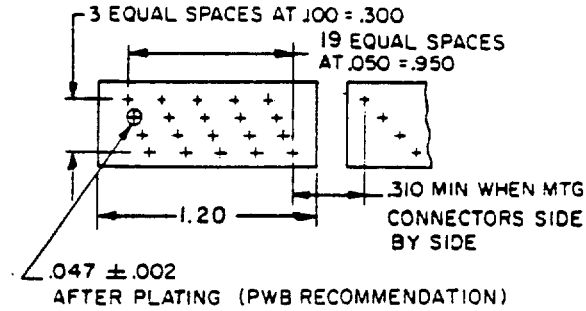
BOTTOM VIEWS

FIGURE 1. Dimensions and configuration - Continued.

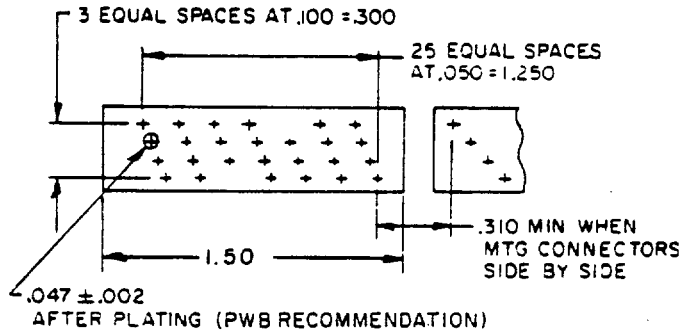
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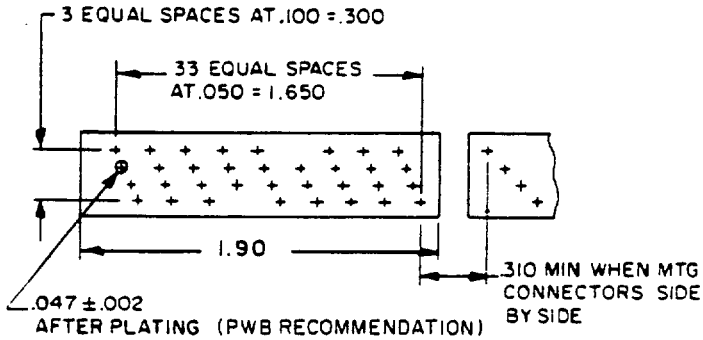
M83503/23-01
AND
M83503/23-02



M83503/23-03
AND
M83503/23-04



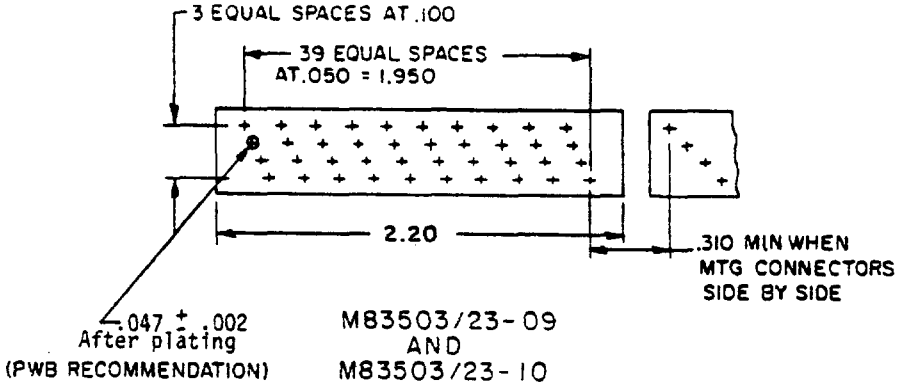
M83503/23-05
AND
M83503/23-06



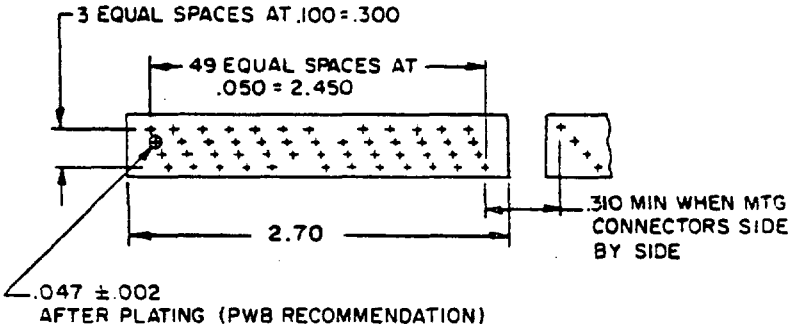
M83503/23-07
AND
M83503/23-08

FIGURE 1. Dimensions and configuration - Continued.

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M83503/23-09
AND
M83503/23-10



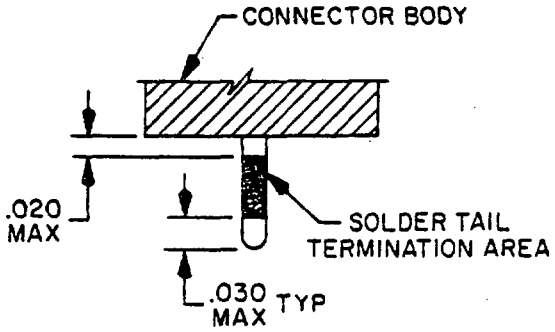
M83503/23-11
AND
M83503/23-12

Part number and characteristics.

Military part number	Number of positions	Dimensions	
		A Max	B ±.010 (.25)
M83503/23-01	10	.72 (18.3)	.100 (2.54)
M83503/23-02	10	.72 (18.3)	.156 (3.96)
M83503/23-03	20	1.22 (31.0)	.100 (2.54)
M83503/23-04	20	1.22 (31.0)	.156 (3.96)
M83503/23-05	26	1.52 (38.6)	.100 (2.54)
M83503/23-06	26	1.52 (38.6)	.156 (3.96)
M83503/23-07	34	1.92 (48.3)	.100 (2.54)
M83503/23-08	34	1.92 (48.3)	.156 (3.96)
M83503/23-09	40	2.22 (56.4)	.100 (2.54)
M83503/23-10	40	2.22 (56.4)	.156 (3.96)
M83503/23-11	50	2.72 (69.1)	.100 (2.54)
M83503/23-12	50	2.72 (69.1)	.156 (3.96)

FIGURE 1. Dimensions and configuration - Continued.

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DETAIL-A
FINISH LOCALIZATION AREA

INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.100	2.54	1.250	31.75
.010	.25	.200	5.08	1.50	38.1
.013	.33	.255	6.48	1.650	41.91
.015	.38	.300	7.62	1.90	48.3
.020	.51	.310	7.87	1.950	49.53
.026	.66	.450	11.43	2.20	55.9
.030	.76	.455	11.56	2.450	62.23
.047	1.19	.70	17.8	2.70	68.6
.050	1.27	.950	24.13		
.070	1.78	1.20	30.5		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.02 (.5 mm) for two place decimals.
4. Method of connection of cover to body optional but shall meet the requirements of MIL-C-83503 and this specification sheet.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Temperature range: -55°C to +120°C.

Current rating: 1 ampere.

Materials:

Contacts: Phosphor bronze, grade A, per QQ-B-750 or beryllium copper per QQ-C-530, or QQ-C-533.

Plating: Tin-lead in the termination area.

Housing: Glass filled polyester per MIL-M-24519, type GPT-15F, GPT-20F or GPT-30F.

Wire accommodation: 28 AWG stranded on .050 spacing between conductor centerlines.

Voltage rating: 300 V rms.

Test wire size: 28 AWG stranded.

Preconditioning: Not applicable.

Dielectric withstanding voltage (at sea level): 500 V rms, minimum.

(at altitude): 200 V rms, minimum.

Contact retention: Not applicable.

Contact resistance: 50 milliohms, maximum.

Mating and unmating force: Not applicable.

Durability: Not applicable.

Part number: M83503/23-(dash number from figure 1).

Assembly tool: M22520/32.

Polarization: Not required.

Marking: Part number shall be ink stamped or molded raised numbers on connector cover.

Interface: Round conductor flat cable, .050 centers to .062 and .093 to .125 thick PW boards.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 85

(Project 5935-F159-4)

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